

A finite-dimensional approximation case of Cornulier–Tessera’s APFD question

FA Banach run fa_banach_001

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Source question

The source is Yves Cornulier and Romain Tessera, *On the vanishing of reduced 1-cohomology for Banach representations*, arXiv:1711.05566. On PDF page 19, Questions 4.6 and 4.7 ask whether Property $\mathcal{AP}_{\text{fd}}$ holds for arbitrary locally compact groups, and more concretely:

Let G be a locally compact group and V an almost periodic Banach G -module. Consider $b \in Z^1(G, V) \setminus \overline{B^1(G, V)}$. Does there exist a G -submodule of finite codimension $W \subset V$ such that the image of b in $Z^1(G, V/W)$ is unbounded?

In view of Proposition [4.3](#), a positive answer would follow from a positive answer to:

Question 4.6. Does Property $\mathcal{AP}_{\text{fd}}$ hold for an arbitrary locally compact group G ?

In turn, a positive answer would result from the following less restrictive question:

Question 4.7. Let G be a locally compact group and V an almost periodic Banach G -module. Consider $b \in Z^1(G, V) \setminus \overline{B^1(G, V)}$. Does there exist a G -submodule of finite codimension $W \subset V$ such that the image of b in $Z^1(G, V/W)$ is unbounded?

For instance, the answer is positive in the case of unitary Hilbert G -modules. See [§4.6](#) for more on Property $\mathcal{AP}_{\text{fd}}$.

Result

The following gives a positive answer for the standard class of almost periodic modules which can be approximated by equivariant finite-dimensional projections.

Theorem 1. *Let G be a locally compact group and let V be an almost periodic Banach G -module. Suppose that there is a net (P_i) of bounded linear projections on V such that:*

1. *each P_i commutes with the G -action;*
2. *each range $F_i = P_i V$ is finite-dimensional;*

3. $P_i v \rightarrow v$ in norm for every $v \in V$.

If $b \in Z^1(G, V)$ is not an almost coboundary, then for some i the projected cocycle $P_i b \in Z^1(G, F_i)$ is unbounded. Consequently, with $W_i = \ker P_i$, the image of b in $Z^1(G, V/W_i)$ is unbounded and W_i is a finite-codimensional G -submodule.

Proof intuition

The obstruction in Question 4.7 is whether a nonzero reduced cohomology class in an almost periodic module must already be visible at finite dimension. If finite-dimensional equivariant projections approximate the identity, then the answer is yes. A bounded finite-dimensional affine action with precompact linear part has a fixed point, so every bounded projected cocycle is an actual coboundary. If all finite-dimensional projections were bounded, their transfer vectors would give almost fixed points for the original affine action, because the discarded part $(I - P_i)b$ is uniformly small on compact subsets of G .

Proof

We first record the elementary finite-dimensional fact used in the argument.

Lemma 1. *Let F be a finite-dimensional Banach G -module whose linear action has precompact image in $\text{GL}(F)$. If $c \in Z^1(G, F)$ is bounded, then c is a coboundary.*

Proof. Let $\alpha(g)x = \pi(g)x + c(g)$ be the corresponding affine action on F . Since $c(G)$ is bounded and $\pi(G)$ is precompact, $\alpha(G)$ has compact closure in the finite-dimensional affine group. This closure is a compact group of affine maps. Averaging any point of F over this compact group with Haar probability measure gives a fixed point x_0 . Thus

$$\pi(g)x_0 + c(g) = x_0$$

for all $g \in G$, i.e. $c(g) = x_0 - \pi(g)x_0$. □

We prove Theorem 1 by contraposition. Assume that every $P_i b$ is bounded. By Lemma 1, for each i there exists $x_i \in F_i$ such that

$$P_i b(g) = x_i - \pi(g)x_i \quad (g \in G).$$

Fix a compact subset $K \subset G$ and $\varepsilon > 0$. Since b is continuous, $b(K)$ is a compact subset of V . The pointwise convergence $P_i \rightarrow I$ and the uniform boundedness principle imply that the net (P_i) is uniformly bounded; hence $P_i \rightarrow I$ uniformly on compact subsets of V . Therefore, for some i ,

$$\sup_{g \in K} \|(I - P_i)b(g)\| < \varepsilon.$$

For this i and every $g \in K$,

$$\pi(g)x_i + b(g) - x_i = b(g) - P_i b(g) = (I - P_i)b(g),$$

so x_i is an ε -almost fixed point for the affine action associated to b on the compact set K . Since K and ε were arbitrary, b is an almost coboundary. This contradicts the hypothesis. Hence some $P_i b$ is unbounded. Since P_i is equivariant, $W_i = \ker P_i$ is a closed G -submodule of finite codimension and V/W_i identifies with F_i . □

Consequences

Corollary 1. *Let V be an almost periodic Banach G -module with an equivariant finite-dimensional decomposition: V is the closed span of finite-dimensional G -submodules (E_n) and the coordinate partial-sum projections $P_N : \overline{\bigoplus E_n} \rightarrow E_1 + \cdots + E_N$ are uniformly bounded and converge strongly to the identity. Then Question 4.7 has a positive answer for cocycles into V .*

Proof. The partial-sum projections satisfy the hypotheses of Theorem 1. □

Remark 1. *This includes the usual c_0 - and ℓ_p -sums, $1 \leq p < \infty$, of finite-dimensional G -modules, with the coordinatewise action. Thus a counterexample to Question 4.7 cannot be obtained merely by taking a diagonal direct sum of finite-dimensional almost periodic representations with the standard coordinate projections.*

Limitations and novelty check

This packet does not prove Property $\mathcal{AP}_{\text{fd}}$ for all locally compact groups. The extra hypothesis is exactly a controlled finite-dimensional approximation of the identity by equivariant projections. General Banach representations of compact groups need not come equipped with such projections, even though finite-dimensional submodules are dense in the almost periodic part.

Run-local cheap indexes were searched for arXiv:1711.05566 and for the terms WAPFD, WAPAP, APFD, and **almost periodic Banach**. No prior run packet for this target was found. A targeted web search for the exact WAPFD/WAPAP/APFD wording did not reveal an obvious later answer to Questions 4.6 or 4.7.

References

1. Yves Cornulier and Romain Tessera, *On the vanishing of reduced 1-cohomology for Banach representations*, arXiv:1711.05566.